

Practice - 02

ADT: Faculty

Write a class named **Faculty** that has the following:

1. The class should have following **five private data members**.
 - An integer named **id** that holds the faculty's identification number. The value of identification number should fall in between 1001 to 1065 both inclusive, 0 otherwise.
 - A string named **firstName** that holds the faculty's first name.
 - A string named **lastName** that holds the faculty's last name.
 - A string named **qualification** that holds the faculty's qualification in which he/she mastered.
 - A float named **salary** that holds the faculty's current monthly salary. The value of **salary** must fall in between 20000.00 and 100000.00 both inclusive, -1 otherwise.
2. Provide the implementation of mutators for all the data members (**id**, **firstName**, **lastName**, **qualification** and **salary**) of the class.
3. Provide the implementation of accessors for all the data members (**id**, **firstName**, **lastName**, **qualification** and **salary**) of the class.
4. Provide the implementation of following **constructors** and a **destructor**.
 - The constructor should accept the faculty's **identification number**, **first name**, **last name**, and **qualification** as arguments. These values should be assigned to the object's appropriate member variables. The constructor should also assign **-1** to the **salary** member variable.
 - The constructor should accept the faculty's **identification number**, **first name** and **qualification** as arguments. These values should be assigned to the object's appropriate member variables. The constructor should also assign empty string ("") to **last name** and **-1** to the **salary** member variable.
 - The constructor should accept the faculty's **identification number**, **first name**, **last name**, **qualification**, and **salary** as arguments. These values should be assigned to the object's appropriate member variables.
 - A copy constructor to initialize a faculty's object with already existing object.
 - A destructor that does nothing except displaying a simple message "Destructor executed..." on the screen.
5. Provide the implementation of following member functions.
 - **set** method accepts faculty's **identification number**, **first name**, **last name**, **qualification**, and **salary** as arguments and assigns them to the appropriate member variables.
 - **read** method to initialize the data of a faculty taken from the user through the console.
 - **write** method to display the information of a particular faculty on the console.
 - **isPostGraduate** method should return *true* if the faculty's qualification is post graduated. i.e., a faculty having MS or Phd. degree(s), *false* otherwise.
 - **getTakeHomeSalary** method should provide the facility to calculate and return the take home salary i.e., $salary - tax$ where $tax = 7\%$ of actual salary only if the value of salary is greater than or equal to 20000.00, otherwise return **-1**.
 - **isJoined** method should return *true* if the faculty's **salary** is greater than or equal to 20000.00, *false* otherwise.
6. In the **main** function, create instances of the **Faculty** class and demonstrate the functionality of each function clearly.

